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Bilateral Gluteal Tuberous Xanthomas in a Young Male with Suspected Familial Hypercholesterolemia and Severe Aortic Stenosis: A Case Report

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Abstract :

Tuberous xanthomas are rare skin manifestations of severe lipid metabolism disorders, most commonly familial hypercholesterolemia (FH). Their presence in young individuals should prompt evaluation for premature cardiovascular and valvular disease. We report a 19-year-old male with multiple swellings since childhood, including painful bilateral gluteal masses, diagnosed as tuberous xanthomas with severe aortic stenosis. Surgical excision was performed, and histopathology confirmed the diagnosis; recovery was uneventful. This case underscores the importance of recognizing xanthomas as markers of inherited dyslipidemia and serious cardiovascular risk, highlighting the need for early diagnosis, aggressive lipid-lowering therapy, and multidisciplinary care.

Keywords : Tuberous xanthoma, homozygous familial hypercholesterolemia, familial hypercholesterolemia, aortic stenosis, dyslipidemia, premature cardiovascular disease

Introduction:

Xanthomas are localized accumulations of lipid laden macrophages within the skin, tendons, fascia, or subcutaneous tissues and are clinically significant because they frequently indicate underlying disorders of lipoprotein metabolism. Histologically, they are characterized by collections of cholesterol rich foam cells formed through excessive uptake and oxidative modification of low-density lipoprotein (LDL) cholesterol by macrophages.

Among the various clinical subtypes, tuberous & tendinous xanthomas are strongly associated with familial hypercholesterolemia (FH), particularly severe forms such as homozygous familial hypercholesterolemia (HoFH). In contrast, eruptive xanthomas are more commonly linked to severe hypertriglyceridemia. Tuberous xanthomas typically present as firm yellow-orange nodules or plaques over pressure bearing areas including the elbows, knees, hands,

feet, and buttocks. Although most lesions are small, giant tuberous xanthomas involving the gluteal region have been rarely described in pediatric and young adult patients with familial hypercholesterolemia. Their appearance in early life is highly suggestive of inherited dyslipidemia and warrants immediate cardiovascular evaluation. Familial hypercholesterolemia is an autosomal dominant genetic disorder characterized by impaired LDL receptor function, markedly elevated serum LDL cholesterol levels from birth. Persistent hypercholesterolemia accelerates atherosclerosis & predisposes affected individuals to premature coronary artery disease, peripheral vascular disease, valvular abnormalities, including aortic stenosis and supraaortic stenosis. Studies have demonstrated that aortic valve involvement in HoFH significantly increases the risk of premature mortality. Cutaneous xanthomas are increasingly recognized as external markers of systemic atherosclerotic disease. Patients with tendinous or tuberous xanthomas have been shown to possess substantially higher cardiovascular risk independent of traditional risk factors. Recognition of these lesions therefore provides an important opportunity for early diagnosis and intervention. We present a rare case of bilateral gluteal tuberous xanthomas in a 19-year-old male with suspected familial hypercholesterolemia and severe aortic stenosis who underwent successful surgical excision.

Case Presentation:

A 19-year-old male presented with multiple swellings distributed over the body since childhood, associated with pain and discomfort over bilateral gluteal swellings. The patient denied fever, vomiting, bowel disturbances, weight loss, jaundice, or other constitutional symptoms. His past medical history was significant for a road traffic accident four years earlier that resulted in an ankle fracture managed surgically with nailing and plating. He had previously undergone excision biopsy of similar swellings over both elbows in 2015 and the postauricular region in 2021. Personal history revealed alcohol consumption, smoking, and tobacco chewing. Family history was notable for cardiovascular disease in his father. On examination, the patient was conscious, alert, and oriented. His pulse rate was 90 beats/minute, blood pressure was 120/80 mmHg, oxygen saturation was 99% on room air, and he was afebrile. Systemic examination was unremarkable. Local examination revealed bilateral pedunculated gluteal swellings with mild tenderness. There was no evidence of erythema, discharge, ulceration, or secondary infection. The longstanding history of multiple xanthomatous lesions beginning in childhood strongly suggested a severe inherited lipid disorder, most likely familial hypercholesterolemia. The coexistence of severe aortic stenosis at the 19 years of age further raised suspicion for homozygous (or) severe

heterozygous familial hypercholesterolemia with premature cardiovascular involvement. Based on the clinical presentation, a provisional diagnosis of bilateral gluteal tuberous xanthomas secondary to familial hypercholesterolemia was made.

Management:

The patient underwent excision biopsy of bilateral gluteal swellings under general anesthesia on 21 July 2025. Following standard preparation & draping, the left gluteal lesion was excised using sharp dissection and electrocautery. Hemostasis was achieved, and a Romovac drain was placed before layered wound closure. A similar procedure was subsequently performed for the right gluteal lesion with drain placement. Compression dressings were applied, and both specimens were sent for histopathological examination. Histopathological examination demonstrated lipid-laden foam cells with cholesterol deposition consistent with tuberous xanthoma. Postoperatively, the patient was monitored in the intensive care unit and later transferred to the surgical ward on postoperative day one. Oral medications were initiated and postoperative recovery remained uneventful. Drains were removed on postoperative day five. The patient was discharged in stable condition with advice regarding wound care, aggressive lipid control, smoking cessation, dietary modification, regular cardiology follow up.

Pathophysiology and Clinical Spectrum of Xanthomas:

Nature and types:

- Xanthomas are well circumscribed lesions in skin tendons, and fascia composed predominantly of lipid-laden foam cells formed from macrophages after excess LDL uptake and oxidative modification (Žák et al., 2014; Tolba et al., 2024; Wang et al., 2016).

- *Subtype strongly predicts underlying lipid disorder:* Tuberous and tendinous xanthomas → typical of autosomal dominant FH and other rare lipidoses (Žák et al., 2014; Tolba et al., 2024; Putri et al., 2024; Kolansky et al., 2008; Morocco., 2018; Guan et al., 2024)- Eruptive xanthomas → severe hypertriglyceridemia and risk of pancreatitis (Žák et al., 2014; Ibrahim et al., 2023)

Tuberous xanthomas :

- Flat or raised yellow-red nodules, usually 3 mm–several cm, over pressure/trauma areas such as elbows, knees, hands/feet, and buttocks (Žák et al., 2014; Tolba et al., 2024; Kolansky et al., 2008; Sharanya et al., 2024; Morocco., 2018; Dung et al., 2021; Guan et al., 2024).

- Usually <2 cm; “giant” or very large lesions (up to 7–8 cm) are rare but repeatedly reported in pediatric and young FH patients, including buttock involvement (Tolba et al., 2024; Putri et al., 2024; Kolansky et al., 2008; Sharanya et al., 2024; Morocco., 2018; Dung et al., 2021; Iqbal et al., 2024; Guan et al., 2024).

Xanthomas as Cardiovascular Risk Markers:

- *Xanthoma formation parallels early atherogenesis*; xanthomas signal congenital or acquired dyslipidemias and increased cardiovascular risk (Žák et al., 2014; Tolba et al., 2024; Tromp et al., 2022; Vargas-Flores et al., 2016).
- In FH, tendinous xanthomas are associated with a 2–4-fold higher risk of cardiovascular disease, independent of mutation type (Žák et al., 2014).

- Xanthelasma and other xanthomas correlate with athero-thrombotic disease & reduced lifespan, even beyond conventional risk factors (Žák et al., 2014; Mustafa & Kumar, 2025; Prajapati et al., 2024; Vargas-Flores et al., 2016).

Familial Hypercholesterolemia, Aortic Valve Disease and Premature ASCVD:

FH phenotype and timing:

- FH (especially homozygous) causes markedly elevated LDL from birth and leads to premature ASCVD and aortic valvulopathy in childhood and young adulthood (Tolba et al., 2024; Kolansky et al., 2008; Ibrahim et al., 2023; Zhang et al., 2021; Tromp et al., 2022; Iqbal et al., 2024).
- Large international & registry data show HoFH diagnosis often delayed (median ~12 years), with aortic stenosis or ASCVD already present at diagnosis in a substantial minority (Tromp et al., 2022).

Aortic root/valve involvement:

- Children with HoFH frequently exhibit early aortic valve disease: mild aortic insufficiency, increased aortic stiffness, and coronary dilatation before overt systolic dysfunction (Kolansky et al., 2008).
- Supravalvular aortic stenosis (and more broadly aortic stenosis) in HoFH is strongly linked to premature death; SVAS independently increased mortality risk more than fourfold in a pediatric HoFH cohort (Zhang et al., 2021).
- Longitudinal pediatric studies document progression of coronary and aortic valve disease in adolescence despite absence of early symptoms, with many patients ultimately requiring valve or coronary surgery (Patel et al., 2020; Kolansky et al., 2008; Zhang et al., 2021).

Xanthomas as First Presentation of FH With Valvular Disease:

- Multiple pediatric and young adult case reports describe tuberous and/or tendinous xanthomas as the presenting clue to HoFH or severe FH, often associated with aortic stenosis or regurgitation and left ventricular hypertrophy (Tolba et al., 2024; Putri et al., 2024; Kolansky et al., 2008; Sharanya et al., 2024; Morocco., 2018; Dung et al., 2021; Iqbal et al., 2024; Guan et al., 2024).
- Tuberous xanthomas have been specifically reported in FH with aortic regurgitation and with left ventricular outflow tract disease in children and adolescents (Tolba et al., 2024; Putri et al., 2024; Kolansky et al., 2008; Dung et al., 2021).

Management: Surgical Excision Plus Aggressive Lipid Lowering Local management of large gluteal/tuberous xanthomas:

- Xanthomas are non-neoplastic; histology shows foam cells and cholesterol clefts in dermis/subcutis or tendon (Tolba et al., 2024; Putri et al., 2024; Zhou et al., 2024; Morocco., 2018; Wang et al., 2016).
- Small lesions may regress with intensive LDL lowering (statins, apheresis) over months, but large, symptomatic, or disfiguring tuberous xanthomas particularly on buttocks & weight bearing areas are commonly treated with surgical excision (Tolba et al., 2024; Putri et al., 2024; Zhou et al., 2024; Morocco., 2018; Dung et al., 2021; Guan et al., 2024).
- *Case series of multiple large tuberous and tendinous xanthomas (including buttocks) show:*

1. Wide but anatomically respectful elliptic excision (parallel to Langer’s lines)
2. Primary closure or local flap; good healing and short-term absence of recurrence (Tolba et al., 2024; Zhou et al., 2024; Morocco., 2018; Dung et al., 2021; Guan et al., 2024).

- Recurrence rates after excision can reach 25–39%, so surgery should be combined with systemic lipid lowering therapy to reduce recurrence and cardiovascular risk (Tolba et al., 2024; Zhou et al., 2024; Morocco., 2018; Dung et al., 2021; Guan et al., 2024).

Systemic therapy and multidisciplinary follow up:

- Larger HoFH registries show better outcomes and later first major cardiovascular events in patients receiving multi-drug regimens and apheresis, particularly in high income settings (Tromp et al., 2022).

- In FH and HoFH cohorts, early and aggressive LDL lowering (high-intensity statins ± ezetimibe, PCSK9 inhibitors, and lipoprotein apheresis where available) is crucial to delay ASCVD, aortic valve disease, and improve survival (Tolba et al., 2024; Patel et al., 2020; Kolansky et al., 2008; Zhang et al., 2021; Tromp et al., 2022; Iqbal et al., 2024).
- Multidisciplinary management dermatology, cardiology (including serial echocardiography for aortic stenosis), lipidology, nutrition, and when needed cardiac surgery is repeatedly emphasized in pediatric and young adult FH case reports (Putri et al., 2024; Patel et al., 2020; Kolansky et al., 2008; Tromp et al., 2022; Iqbal et al., 2024; Guan et al., 2024).

Sites, Age, and Diagnostic Implications Relevant to The Case:

Feature	Typical in FH / HoFH	Relevance to your 19-year-old case
Onset of xanthomas before age 10–20	Common, especially in HoFH	Longstanding lesions since childhood support severe inherited dyslipidemia
Tuberous xanthomas on buttocks	Well-described but uncommon	Bilateral gluteal masses fit classical distribution
Association with aortic stenosis / regurgitation	Documented in pediatric FH	Severe aortic stenosis at 19 strongly suggests HoFH or severe FH
Benefit of combining surgery + lipid lowering	Repeated across case reports	Justifies the excision plus post-op lipid management strategy

Figure 1: Clinical patterns linking xanthomas, FH, and aortic disease

Discussion:

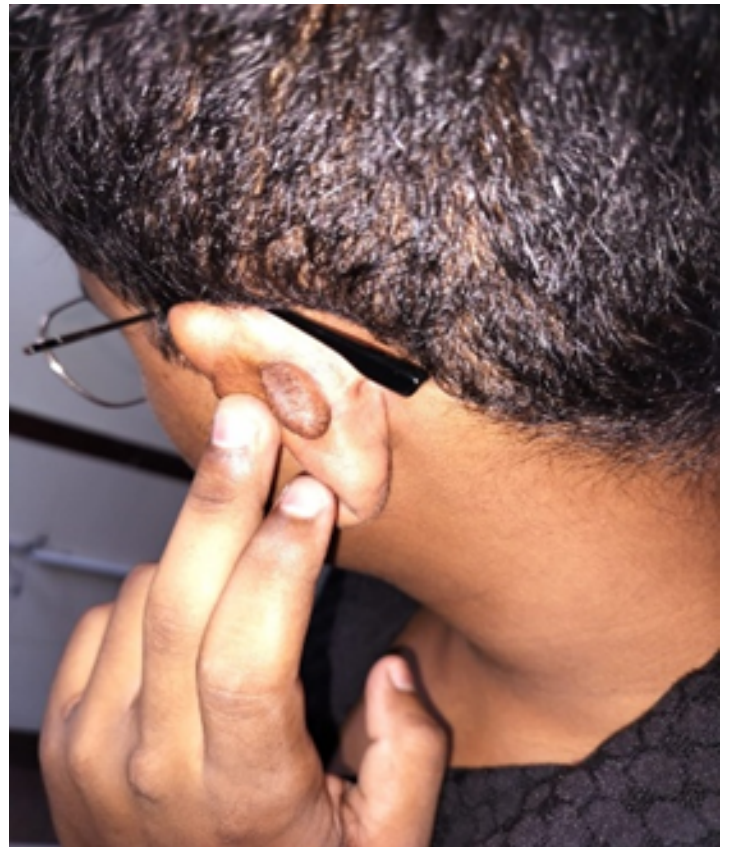
Tuberous xanthomas are nodular lesions resulting from dermal & subcutaneous deposition of cholesterol rich foam cells. They most commonly occur over extensor surfaces and pressure-bearing regions such as the elbows, knees, Achilles tendons, and buttocks. Bilateral gluteal involvement, as observed in the present case, is relatively uncommon but has been repeatedly reported in patients with severe familial hypercholesterolemia. The occurrence of multiple xanthomas since childhood strongly suggests inherited dyslipidemia, particularly homozygous familial hypercholesterolemia. Patients with HoFH develop markedly elevated LDL cholesterol

levels from birth and are predisposed to accelerated atherosclerosis, premature coronary artery disease, valvular heart disease during childhood or early adulthood. Delayed diagnosis remains a significant concern worldwide, & many patients already demonstrate advanced cardiovascular involvement at the time of initial presentation. An important feature in this patient was the coexistence of severe aortic stenosis at a young age. Lipid deposition, chronic inflammation, and accelerated calcific degeneration of the aortic valve are recognized consequences of longstanding hypercholesterolemia. Pediatric and adolescent studies have demonstrated that patients with HoFH may develop early aortic valvulopathy, coronary artery disease, supraaortic stenosis, and left ventricular hypertrophy even before the onset of overt symptoms. Supraaortic and valvular aortic stenosis in this setting are associated with markedly increased mortality risk. The pathophysiological mechanisms underlying xanthoma formation closely parallel those involved in atherosclerosis. Excess circulating LDL undergoes oxidative modification & subsequently engulfed by macrophages, resulting in foam cell accumulation within tissues. Consequently, cutaneous xanthomas should not be regarded as isolated dermatologic lesions but rather as visible indicators of severe systemic lipid dysregulation and heightened cardiovascular risk. Management of tuberous xanthomas primarily focuses on aggressive correction of dyslipidemia through lifestyle modification& pharmacologic LDL-lowering therapy. High-intensity statins, ezetimibe, PCSK9 inhibitors, & lipoprotein apheresis may be required, particularly in severe familial hypercholesterolemia. Early institution of combination lipid-lowering therapy has been shown to delay progression of atherosclerotic cardiovascular disease and improve survival. Surgical excision is indicated for symptomatic lesions, cosmetic disfigurement, ulceration, recurrent trauma, restriction of movement, or diagnostic confirmation. In the present case, surgical excision provided symptomatic relief and histopathological confirmation. Previous reports have demonstrated favorable outcomes following excision of large gluteal and tendinous xanthomas, although recurrence may occur if systemic lipid abnormalities remain uncontrolled. This case emphasizes the importance of a multi-disciplinary approach involving surgeons, dermatologists, cardiologists, lipid specialists, and nutritionists. Young patients presenting with xanthomas should undergo comprehensive metabolic evaluation and echocardiographic assessment to identify potentially life-threatening cardiovascular complications at an early stage.

Conclusion:

For journal submission, emphasize that bilateral gluteal tuberous xanthomas in a teenager are a striking phenotypic

marker of severe FH, tightly linked to early and potentially fatal aortic valve and coronary disease. Highlight the characteristic distribution and the histology of tuberous xanthomas, their strong association with HoFH, and the evidence that supra-valvular/aortic stenosis in this setting confers markedly increased risk of premature death. The case supports the message that prompt recognition of xanthomas, urgent cardiologic evaluation (especially echocardiographic assessment of the aortic valve), aggressive LDL-lowering, and the selected surgical excision of large symptomatic lesions are essential to improving quality of life and long-term outcomes in young patients with FH. Tuberous xanthomas in children and young adults are important clinical indicators of severe underlying dyslipidemia, particularly familial hypercholesterolemia. Bilateral gluteal tuberous xanthomas in a teenager represent a striking phenotypic manifestation of severe familial hypercholesterolemia and should prompt urgent cardiovascular evaluation. This case highlights and the close association between cutaneous xanthomas and the early severe aortic stenosis, emphasizing that xanthomas are not merely cosmetic lesions but external markers of potentially fatal systemic disease. Early recognition, and aggressive lipid-lowering therapy, regular echocardiographic surveillance, multidisciplinary management, and the selective surgical excision of symptomatic lesions are essential to improve quality of life & reduce long-term cardiovascular morbidity and mortality.





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